Controlling Table Placement in R Markdown PDF Reports

Practical strategies for reliable table and figure positioning in LaTeX-rendered PDF output

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Figure 1: R Markdown provides a bridge between R analysis code and professionally formatted PDF reports via LaTeX.

The gap between what you write and where it appears in the PDF is controlled by LaTeX's float algorithm.

1 Introduction

[YOUR CONTENT HERE – 2-3 paragraphs. Open with an "I didn't really know…" hook about table placement frustration. Reference the Stack Overflow quote from the original post:

> "I have read what feels like 100 articles and threads on the topic and tried so many things I have lost count of it all but am still stuck."

Describe the universal frustration with LaTeX float placement in R Markdown PDFs. Transition to what this post will cover.]

1.1 Motivations

[YOUR CONTENT HERE – 4-6 bullet points. Personal and specific reasons. Examples: tables appearing pages away from the text referencing them, figures jumping to the end of the document, client reports with misplaced tables, etc.]

1.2 Objectives

1. [YOUR CONTENT HERE – specific, measurable]
2. [YOUR CONTENT HERE]
3. [YOUR CONTENT HERE]
4. [YOUR CONTENT HERE]

I am documenting my learning process here. If you spot errors or have better approaches, please let me know.

The real test is printing the PDF and seeing where everything landed.

2 Prerequisites and Setup

[YOUR CONTENT HERE – 1-2 paragraphs. What readers need: R, rmarkdown, knitr, kableExtra, a LaTeX distribution (tinytex or texlive). Show library loading.]

```
library(knitr)
library(kableExtra)
library(tidyverse)
```

3 What is LaTeX Float Placement?

[YOUR CONTENT HERE – 1-2 paragraphs in plain language. Explain that LaTeX treats tables and figures as "floats" that it positions according to an algorithm, not necessarily where you wrote them in the source. Explain the float specifiers: h (here), t (top), b (bottom), p (page), H (exactly here). Use an analogy – e.g., LaTeX is like a newspaper layout editor that moves your photo to wherever it fits best on the page.]

4 Getting Started: The Default Behaviour

[YOUR CONTENT HERE – Show a simple R Markdown example that generates a table with kable(). Render to PDF and observe where the table actually appears. Demonstrate the default float behaviour. Follow CODE -> OUTPUT -> INTERPRETATION.]

```
# [YOUR CODE HERE -- simple kable example]
```

[YOUR INTERPRETATION HERE – describe where the table actually appeared vs where you expected it.]

Figure 2: A document layout showing the relationship between source code position and rendered output position.

Understanding the gap between source position and rendered position is the first step.

4.1 The Float Specifier Approach

[YOUR CONTENT HERE – Show how to use kableExtra's kable_styling(latex_options = …) and the float specifier options. Demonstrate [H] with the float package, [!ht], and other combinations.]

```
# [YOUR CODE HERE -- kableExtra with float options]
```

[YOUR INTERPRETATION HERE]

4.2 The longtable Approach

[YOUR CONTENT HERE – Show how longtable avoids the float problem entirely for tables that span pages. Demonstrate with a longer table.]

```
# [YOUR CODE HERE -- longtable example]
```

[YOUR INTERPRETATION HERE]

5 The LaTeX Preamble Approach

[YOUR CONTENT HERE – Show how to modify the LaTeX preamble in the YAML header to globally control float behaviour. Include the flafter package, the float package, and custom placement commands.]

```
# [YOUR YAML HERE -- preamble example]
```

[YOUR INTERPRETATION HERE]

6 Checking Our Work

[YOUR CONTENT HERE – Compare multiple approaches side by side. Show the same table rendered with different float strategies and describe the results. Which approach worked best for which situation?]

```
# [YOUR CODE HERE -- comparison examples]
```

[YOUR INTERPRETATION HERE]

6.1 Things to Watch Out For

1. **[YOUR GOTCHA HERE]** – [explanation]
2. **[YOUR GOTCHA HERE]** – [explanation]
3. **[YOUR GOTCHA HERE]** – [explanation]
4. **[YOUR GOTCHA HERE]** – [explanation]
5. **[YOUR GOTCHA HERE]** – [explanation]

Figure 3: A clean desk with a finished PDF report, representing the satisfaction of correctly placed tables and figures.

When every table lands where you intended, the report is ready.

6.2 Lessons Learnt

6.2.1 Conceptual Understanding

- [YOUR CONTENT HERE – what you learnt about LaTeX float algorithms and PDF rendering]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]

6.2.2 Technical Skills

- [YOUR CONTENT HERE – tools/packages mastered: kableExtra, float package, preamble customisation, etc.]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]

6.2.3 Gotchas and Pitfalls

- [YOUR CONTENT HERE – practical warnings for future R Markdown PDF authors]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]

6.3 Limitations

- [YOUR CONTENT HERE – 5-6 bullets. Cover: LaTeX float algorithm complexity, HTML output behaves differently, Quarto vs R Markdown differences, page geometry constraints, etc.]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]

6.4 Opportunities for Improvement

1. [YOUR CONTENT HERE – extensions: Quarto equivalent techniques, automated placement checking, custom LaTeX templates, etc.]
2. [YOUR CONTENT HERE]
3. [YOUR CONTENT HERE]
4. [YOUR CONTENT HERE]
5. [YOUR CONTENT HERE]
6. [YOUR CONTENT HERE]

7 Wrapping Up

[YOUR CONTENT HERE – 2-3 paragraphs. First: recap the main strategies. Second: what you learnt personally. Third: advice for readers.]

Main takeaways:

- [YOUR CONTENT HERE – 3-4 specific results or strategies]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]
- [YOUR CONTENT HERE]

8 See Also

Related posts:

- [YOUR RELATED POST 1]
- [YOUR RELATED POST 2]

Key resources:

- kableExtra: Construct Complex Table with kable and Pipe Syntax – The primary table formatting package for R
- R Markdown Cookbook: Tables – Yihui Xie's comprehensive reference
- LaTeX Float Placement (Overleaf) – Clear explanation of float specifiers
- Stack Overflow: Table Placement in R Markdown – The community discussion that prompted this post
- tinytex: A Lightweight LaTeX Distribution – Minimal LaTeX for R users

9 Reproducibility

[YOUR CONTENT HERE – environment description, R version, LaTeX distribution, key packages.]

File	Purpose
[YOUR FILES HERE]	[descriptions]

```
# To reproduce:
# [YOUR PIPELINE COMMANDS HERE]
```

10 Let's Connect

- **GitHub:** rgt47
- **Twitter/X:** @rgt47
- **LinkedIn:** Ronald Glenn Thomas
- **Email:** rgtlab.org/contact

I would enjoy hearing from you if:

- You spot an error or a better approach to any of the code in this post.
- You have suggestions for topics you would like to see covered.
- You want to discuss R programming, data science, or reproducible research.
- You have questions about anything in this tutorial.
- You just want to say hello and connect.